ISBMM104 Data Analysis – UNamur

Hands-on session – Module 8 (PREDICTION MODELLING 1)

AVAILABLE MATERIALS

- dataset *ISBMM104_PredMod.csv*
- data dictionary *ISBMM104_PredMod_ddict.csv*
- R cheat-sheet

OBJECTIVE

Using the dataset **ISBMM104_PredMod.csv**, develop, validate, and interpret a prediction model estimating the risk of **severe hypoglycaemia** among older adults (65 years or more) living with type 2 diabetes.

OUTCOMES

Severe hypoglycemia within one year ("severe_hypoglycaemia").

CANDIDATE PREDICTORS

Variable	Description
age	Age (years)
sex	Sex
deprivation_score	Socioeconomic deprivation score
diabetes_duration	Duration of diabetes
hba1c	HbA1c (%)
bmi	Body mass index
sulfonylurea_use	Sulfonylurea use
glp1ra_use	GLP-1 receptor agonist use
sglt2i_use	SGLT2 inhibitor use
basal_insulin_use	Basal insulin use
basal_prandial_insulin_use	Basal-prandial insulin use
egfr	Kidney function
frailty_index	Frailty index
dementia	Dementia
falls_history	Previous falls
previous_hypoglycaemia	Previous hypoglycaemia
polypharmacy	Number of medications
weight_loss	Recent weight loss

ANALYSIS AND TASKS

Part 1 – Explore the Dataset

Task 1

Import the dataset into R.

Report:

- Number of patients
- Number of outcome events
- Outcome prevalence (%)

Task 2

Explore all candidate predictors. For each predictor:

- Determine variable type
- Identify missing values
- Produce descriptive statistics

Part 2 – Complete-case dataset

In this first analysis, use only patients with complete data.

Task 3

Create a complete-case dataset.

Report:

- Number of patients excluded
- Number of patients remaining

Task 4

Use a penalised logistic regression model to predict severe hypoglycaemia.

Use: `cv.glmnet()` with `family = "binomial"` and `alpha = 1`.

Task 5

Extract and compare the model coefficients for:

- Weak penalization: lambda: lambda.min / 5
- Optimal penalisation: lambda = lambda.min
- Strong penalisation: lambda = lambda.1se
- Very strong penalisation: lambda = lambda.1se * 3

For each model, report:

- Number of selected predictors
- Selected predictors
- Coefficient estimates

Present your results in a table.

Task 6

Report the final model (using either lambda.min or lambda.1se).

Present:

- Selected predictors
- Estimated coefficients
- Predictors excluded by the penalisation procedure

Questions:

1. Which predictors remained in the final model?
2. Which predictors were removed?
3. Do the selected predictors make clinical sense?
4. What are the advantages and disadvantages of using lambda.min?
5. What are the advantages and disadvantages of using lambda.1se?
6. Which model would you choose for clinical implementation, and why?

Part 3 – Apparent Performance

Evaluate the predictive performance on the same dataset.

Task 7 – Discrimination

Calculate:

- AUROC

Interpret the value according to the categories discussed during the lecture.

Questions:

- Is discrimination poor, acceptable, good, or excellent?
- What does the AUROC mean clinically?

Task 8 – Calibration

Construct a calibration plot.

Hints:

1. Obtain predicted probabilities.
2. Divide patients into groups according to predicted risk.
3. Compare observed and predicted risks.

Questions:

- Does the model systematically overestimate risk?
- Does the model systematically underestimate risk?

Part 4 – Internal Validation

Apparent performance is optimistic. We therefore need to perform internal validation.

Task 9

Use bootstrap validation.

Perform: 500 bootstrap samples

For each bootstrap sample:

1. Refit the model.
2. Calculate performance in the bootstrap sample.
3. Calculate performance in the original sample.
4. Estimate optimism.

Compute: Optimism-corrected AUROC

Task 10

Compare Apparent and bootstrap-corrected AUROC

Questions:

- Is optimism large or small?
- What does this tell you about overfitting?